Diretrizes de Cuidados Intermediários

LOC: LOC-002 (ISC)

MCG Health -
Cuidados Hospitalares e Cirúrgicos
- 29ª Edição

Esta diretriz de cuidados da MCG ainda não foi atualizada para refletir as novas orientações e requisitos encontrados na Regra Final do Medicare Advantage e da Parte D de 2024 (CMS-4201-F). Para beneficiários do Medicare Advantage, antes de consultar as diretrizes de cuidados da MCG, os usuários devem primeiro seguir todas as orientações relacionadas a este tópico encontradas nas Determinações de Cobertura Nacional (NCDs) e nas Determinações de Cobertura Local (LCDs) do CMS, bem como em outros estatutos e regulamentos. O conteúdo utilizado nas diretrizes de cuidados da MCG pode ser usado para complementar, mas não substitui, modifica ou revoga os regulamentos existentes do Medicare ou as NCDs ou LCDs aplicáveis.

- Diretrizes de Admissão em Cuidados Intermediários
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Diretrizes de Admissão em Cuidados Intermediários

- A admissão em cuidados intermediários pode ser indicada após tratamento intensivo (por exemplo, como cuidados "transitórios") ou para fornecer um nível de cuidados superior ao da enfermaria geral do hospital (por exemplo, como cuidados "progressivos" na ausência de necessidade de admissão em cuidados intensivos) (ver Diretrizes de Cuidados Intensivos ISC), conforme indicado por **um ou mais** dos seguintes [A](1)(2)(3)(4)(5)(6)(7):
  - Necessidade de monitoramento ou cuidados de nível intermediário, conforme indicado por **um ou mais** dos seguintes critérios:
    - Cuidados que exigem uma proporção de enfermeiro para paciente de 1 para 2 ou 1 para 3 (ex.: tratamento complexo de feridas, paciente pós-operatório de alto risco, fraqueza neuromuscular profunda)(1)
    - Monitoramento frequente (mas menos frequente do que a cada hora), como verificação de sinais vitais, exames vasculares ou neurológicos.(3)(4)
    - Necessidade estável de ventilação mecânica crônica ou de desmame a longo prazo(3)(4)
    - Ventilação não invasiva [B][C](1)(2)(3)(8)
    - Terapia com oxigênio de alto fluxo [D](9)(10)(11)
    - Diálise peritoneal
    - Terapia pulmonar frequente com aspiração endotraqueal (paciente não intubado ou com traqueostomia crônica)
    - Diurese rápida para sobrecarga de fluidos
    - Problemas eletrolíticos, metabólicos ou de outra natureza que exigem exames laboratoriais frequentes ou ajustes no tratamento.(2)(3)(12)(13)(14)(15)
    - Monitoramento da anestesia epidural contínua de alto nível.
    - Monitoramento após ingestão de substâncias tóxicas com risco de comprometimento neurológico, cardíaco ou respiratório.(2)
    - Titulação de agentes vasodilatadores ou antiarrítmicos intravenosos(1)(3)
    - Terapia inotrópica ou vasodilatadora em baixa dose, sem necessidade de titulação frequente.(1)(2)(3)
    - Monitoramento contínuo da oximetria de pulso(3)

- Monitoramento eletroencefalográfico de curto prazo(2)
- Paciente que necessita de monitoramento hospitalar de curto prazo (ex.: cardíaco, local da ferida, perfusão)(4)

Diagnósticos ou procedimentos **cardiológicos , incluindo um ou mais** dos seguintes:

- Angina ou síndrome de dor torácica de origem indeterminada com **um ou mais** dos seguintes sintomas:(16)(17)(18):
  - Pressão arterial sistólica inferior a 110 mmHg
  - Arritmias cardíacas de preocupação imediata
  - Edema pulmonar
- Angina instável de início recente em repouso ou com esforço mínimo.
- Infarto agudo do miocárdio ou suspeita de infarto com estabilidade hemodinâmica (19)
- Paciente de baixo risco com infarto agudo do miocárdio com supradesnivelamento do segmento ST submetido a intervenção coronária percutânea bem-sucedida.(19)(20)
- Arritmia cardíaca com estabilidade hemodinâmica (21)(22)(23)
- Bloqueio atrioventricular com estabilidade hemodinâmica (até a resolução do bloqueio ou implante de marca-passo permanente)
- Emergência hipertensiva (ex.: pressão arterial sistólica superior a 180 mmHg ou pressão arterial diastólica superior a 120 mmHg com lesão de órgãos-alvo) sem necessidade adicional de terapia parenteral (ex.: tratamento parenteral bem-sucedido no pronto-socorro com mudança para medicação oral).(24)(25)
- Insuficiência cardíaca moderada sem hipotensão (26)(27)(28)
- Terapia inotrópica ou vasodilatadora em baixa dose sem necessidade de titulação frequente [E](2)(29)

Diagnósticos ou procedimentos **de Cirurgia Cardiovascular , incluindo um ou mais** dos seguintes:(30)(31)(32):

- Paciente submetido à cirurgia cardíaca, hemodinamicamente estável e com mais de 18 horas de pós-operatório.
- Bypass arterial periférico
- Implante de stent endovascular não cardíaco

Diagnósticos ou procedimentos **de endocrinologia , incluindo um ou mais** dos seguintes:(33):

- Cetoacidose diabética ou estado hiperosmolar sem coma, mas que requer testes e tratamento frequentes (por exemplo, ajustes de insulina).(34)(35)(36)(37)
- Hipoglicemia persistente que requer monitoramento frequente e reposição de glicose.(34)(38)
- Estados de hipertireoidismo ou hipotireoidismo que requerem monitoramento frequente.(39)(40)(41)(42)

Diagnósticos ou procedimentos **gastroenterológicos , incluindo um ou mais** dos seguintes:(33):

- Hemorragia gastrointestinal com estabilidade hemodinâmica (43)(44)(45)(46)(47)(48)(49)
- Hemorragia varicosa recente com estabilidade hemodinâmica e sem evidência de sangramento ativo.(47)(49)(50)
- Monitoramento após procedimentos de derivação transcutânea para sangramento varicoso
- Insuficiência hepática aguda com estabilidade hemodinâmica (51)(52)(53)(54)(55)(56)
- Encefalopatia hepática leve (ex.: estágio 1 (confuso) ou estágio 2 (letárgico))(57)
- Pancreatite aguda com estabilidade hemodinâmica que requer reposição volêmica ou monitoramento frequente de eletrólitos devido a grandes alterações de fluidos.(58)(59)
- Gastroenterite com estabilidade hemodinâmica e perda contínua de fluidos, necessitando de reposição volêmica com grandes volumes e monitoramento frequente de eletrólitos.

Diagnósticos ou procedimentos **de Cirurgia Geral , incluindo pacientes com** estabilidade hemodinâmica pós-operatória que necessitam **de um ou mais** dos seguintes:(1)(2)(3)(60):

- Monitoramento rigoroso no primeiro dia pós-operatório (por exemplo, devido ao risco do procedimento, comorbidades subjacentes, necessidade de exames de sangue frequentes, complicações cirúrgicas ou anestésicas, ou alto risco de complicações pós-operatórias).(2)(61)
- Reposição de fluidos devido a alterações significativas no volume hídrico.
- Tratamento extensivo de feridas
- atendimento a lesões múltiplas complexas
- Necessidade de ventilação não invasiva pós-operatória

Diagnóstico de doenças infecciosas , incluindo a resolução da sepse com estabilidade hemodinâmica.

Diagnósticos ou procedimentos **neurológicos , incluindo um ou mais** dos seguintes:(2)(3)(33)(62):

- Acidente vascular cerebral isquêmico agudo [F](63)(64)
- Ingestão de medicamentos com instabilidade hemodinâmica que requer monitoramento clínico ou laboratorial frequente.(2)(65)

- Necessidade de monitoramento neurológico frequente (por exemplo, a cada 2 horas) além da capacidade da enfermaria médica ou pediátrica (por exemplo, crise convulsiva responsiva à terapia que requer monitoramento de EEG de curto prazo)(2)
- Fraqueza neuromuscular que requer monitorização frequente da capacidade vital pulmonar (ex.: miastenia grave, síndrome de Guillain-Barré) ou administração de ventilação não invasiva.(2)(66)(67)(68)(69)(70)(71)(72)(73)(74)

Diagnósticos ou procedimentos **neurocirúrgicos , incluindo um ou mais** dos seguintes:(2)(75):

- Lesão cerebral traumática aguda (escore na Escala de Coma de Glasgow superior a 9) Calculadora da Escala de Coma de Glasgow
- Qualquer procedimento neurocirúrgico importante que exija monitoramento no primeiro dia pós-operatório.
- Hemorragia subaracnoide pré ou pós-operatória que requer monitoramento de vasoespasmo
- Lesão aguda da medula espinhal após estabilização
- Dreno lombar instalado

Diagnósticos ou procedimentos de **Cirurgia Torácica e Doenças Pulmonares , incluindo um ou mais** dos seguintes:(2)(3):

- Insuficiência respiratória com estabilidade hemodinâmica e **um ou mais** dos seguintes critérios :(1)(33)(76):
  - É necessário monitoramento frequente para detectar possíveis deteriorações.
  - Terapia pulmonar frequente (ex.: aspiração, terapia com broncodilatadores)
  - pressão positiva contínua nas vias aéreas nasais
  - Ventilação não invasiva intermitente ou contínua [G](33)(77)(78)
  - Oxigênio aquecido e umidificado por cânula nasal de alto fluxo(11)
- Ventilação mecânica estável a longo prazo
- Necessidade de ventilação não invasiva [G]
- Asma com dispneia grave ou fadiga que requer monitoramento ou terapia frequente, CPAP ou ventilação não invasiva [G](79)
- DPOC com dispneia grave ou fadiga que requer monitoramento ou tratamento frequentes.(80)
- Cor pulmonale com dispneia grave ou fadiga que requer monitoramento ou tratamento frequentes.(80)
- Hipoventilação associada à obesidade, com **um ou mais** dos seguintes sintomas:(81)(82)(83):
  - Nova necessidade de ventilação não invasiva
  - Necessidade de ajuste de ventilador não invasivo pré-existente com necessidade de monitoramento rigoroso do estado clínico (ex.: monitoramento do estado mental ou hipercapnia).
  - Hipoxemia de início recente que persiste apesar do tratamento em observação.
  - Agravamento da hipercapnia pré-existente (por exemplo, aumento documentado da pressão parcial de dióxido de carbono superior a 5 mmHg (0,7 kPa) em relação ao nível basal da doença, com acidose respiratória (pH inferior a 7,35)).
  - Agravamento da hipoxemia pré-existente (por exemplo, aumento da necessidade de oxigênio suplementar para manter a oxigenação no nível basal) com necessidade de monitoramento rigoroso do estado clínico (por exemplo, monitoramento do estado mental ou agravamento da hipercapnia devido ao uso de oxigênio).
- Embolia pulmonar de risco baixo a intermediário (submaciva de baixo risco) com estabilidade hemodinâmica e evidência de sobrecarga do ventrículo direito (**por** exemplo, por ecocardiograma, angiotomografia computadorizada ou ECG) **ou** biomarcadores cardíacos positivos.(84)(85)
- Fratura traumática de costela com capacidade vital forçada entre 26% e 45% do previsto.(86)(87)
- Monitoramento cardiopulmonar após um Evento Aparentemente Ameaçador à Vida (também conhecido como Evento Breve Resolvido Inexplicável) em lactentes com menos de 1 ano de idade.(88)(89)
- Monitoramento do paciente submetido à toracotomia e do débito do dreno torácico no primeiro dia pós-operatório.
- Paciente com obstrução das vias aéreas, necessitando de aspiração frequente e monitorização respiratória.(2)

Diretrizes de Alta para Cuidados Intermediários

- A alta para cuidados intermediários é indicada quando **um ou mais** dos seguintes critérios estão presentes.(1)(2)(90):
  - O estado fisiológico estabilizou-se de tal forma que não são mais necessários cuidados intermediários, conforme indicado por **TODOS** os seguintes critérios:
    - Não há necessidade de monitoramento frequente, verificação de sinais vitais ou cuidados especiais.

- Interrupção da ventilação assistida [C]
- Não há exigência de enfermagem de nível intermediário (ou seja, paciente sem necessidade de uma proporção de 2 ou 3 pacientes por enfermeiro).

- O estado fisiológico deteriorou-se de tal forma que agora é necessário tratamento em UTI. Consulte as Diretrizes de Terapia Intensiva (ISC) .
- O estado fisiológico deteriorou-se a tal ponto que intervenções ativas que requerem um nível intermediário de cuidados já não estão planeadas.(91)

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Footnotes

[A] The intermediate care guideline should not be used independently but in conjunction with the appropriate Inpatient & Surgical Care or General Recovery Care guideline. [A in Context Link 1]

[B] Noninvasive ventilation may be administered in some intermediate care units depending on unit capabilities, staffing, and policies.(2)(3)(8) [B in Context Link 1]

[C] Ventilatory assistance includes noninvasive methods (eg, high-flow nasal cannula (HFNC), CPAP, BPAP) and invasive methods (intubation with mechanical ventilation). [C in Context Link 1, 2]

[D] A specialty society guideline recommends the use of high-flow nasal oxygen over conventional oxygen therapy for patients with acute hypoxic respiratory failure because it decreases the risk of subsequent intubation and mechanical ventilation.(9) High-flow oxygen is also recommended over conventional oxygen therapy for postextubation hypoxic respiratory failure due to a slightly lower likelihood for reintubation and improved patient comfort.(10) [D in Context Link 1]

[E] Low doses of vasopressor or inotropic agents may be administered in some intermediate care units depending on unit capabilities, staffing, and policies.(29) [E in Context Link 1]

[F] Dedicated intermediate-level stroke care units are recommended for acute ischemic stroke. A unit that incorporates rehabilitation is preferred. Telemetry should include blood pressure, pulse, respirations, and pulse oximetry.(63)(64) [F in Context Link 1]

[G] Continuous noninvasive ventilation may be administered in some intermediate care units depending on unit capabilities, staffing, and policies.(2)(8)(77) [G in Context Link 1, 2, 3]

Definitions

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
  - Coma (ie, not arousable)

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Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
  - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
    - Resuscitated ventricular fibrillation or cardiac arrest
    - Ventricular escape rhythm
    - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)

    - Nonsustained ventricular tachycardia and **1 or more** of the following:
        - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
        - Acute myocarditis
- Unstable cardiac conduction defects, as indicated by **1 or more** of the following(5)(7):
    - Type II second-degree atrioventricular block
    - Third-degree atrioventricular block
    - New-onset left bundle branch block with suspected myocardial ischemia
- Any heart rhythm and **1 or more** of the following(4)(8)(9)(10):
    - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
    - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
- Heart rhythms of concern due to **1 or more** of the following:
    - Hypotension
    - Respiratory distress
    - Association with other significant symptoms (eg, syncope)(8)(9)

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Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
    - Hemodynamic abnormalities at baseline or acceptable for next level of care
    - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
        - Tachycardia absent
        - Hypotension absent
        - No evidence of inadequate perfusion (eg, no myocardial ischemia)
        - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
  - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
  - Low blood pressure, as indicated by **1 or more** of the following:
    - SBP less than 90 mm Hg in adult or child age 10 years or older[A][B]
    - Mean arterial pressure[C] less than 70 mm Hg in adult or child age 10 years or older[A][B] MAP Calculator
    - Decrease in baseline mean arterial pressure of 25% or more,[C] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)[A][B] MAP Calculator
    - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A][B]
    - SBP less than 70 mm Hg in infant 1 to 11 months of age[A][B]

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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of blood pressure requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline blood pressure, medications, and clinical impact. For example, an elderly patient with heart failure may be prescribed medication with the intentional goal of a reduced pressure. If the patient's baseline SBP is 92 mm Hg to 96 mm Hg, absent signs or symptoms of hypotension, an emergency department SBP of 86 mm Hg should not automatically be categorized as hypotensive, even though this is below the cutoff of 90 mm Hg. This would hold for mean arterial pressure (MAP) as well. If the baseline blood pressure is 92 mm Hg systolic and 64 mm Hg diastolic (MAP 73.3 mm Hg), absent signs or symptoms of hypotension, an emergency department blood pressure of 86 mm Hg systolic and 58 mm Hg diastolic (MAP 67.3 mm Hg) may not denote hypotension, even though the MAP is below 70 mm Hg. Conversely, a patient with a baseline blood pressure of 150 mm Hg systolic and 90 mm Hg diastolic (MAP 110 mm Hg), with signs or symptoms of hypotension, can be considered hypotensive even with an emergency department blood pressure of 100 mm Hg systolic and 80 mm Hg diastolic (MAP 86.7 mm Hg). The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term hypotension. Whether a blood pressure below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

C. The mean arterial pressure (MAP) takes into account both SBP and DBP readings.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4)(5):
  - SBP greater than or equal to 90 mm Hg in adult or child age 10 years or older
  - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child age 10 years or older MAP Calculator
  - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure) MAP Calculator
  - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
  - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. The mean arterial pressure (MAP) takes into account both SBP and DBP readings.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
  - Patient without baseline need for supplemental oxygen with oxygen saturation (SpO_2) of less than 90% or arterial blood gas partial pressure of oxygen (PaO_2) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
  - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SpO_2 greater than 89% or PaO_2 greater than 59 mm Hg (7.9 kPa)[A]
  - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

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Footnotes

A. Pulse oximetry (SpO_2) may underestimate arterial saturation (SaO_2), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(3)

B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
  - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
  - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension, which relies upon blood pressure alone.(1)(2)(3)

B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
  - Patient with **1 or more** of the following:
    - Dyspnea (difficulty breathing)
    - Tachypnea
    - Abnormal breathing pattern (eg, chest retractions)
    - Other evidence of difficulty breathing
  - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
    - Hypoxemia
    - Altered mental status

- Other evidence of respiratory compromise (eg, respiratory acidosis)

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Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Cooney KA, editors. Goldman-Cecil Medicine. 27th ed. Elsevier; 2024:32-35.
2. Cico SJ, Caglar D. Care of the pediatric patient. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:1996-2004.e1.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):
  - Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
  - Greater than 22 breaths per minute in child 6 to 12 years of age[A]
  - Greater than 25 breaths per minute in child 3 to 5 years of age[A]
  - Greater than 30 breaths per minute in child 1 or 2 years of age[A]
  - Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
  - Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
  - Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

References

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3. Anderson CC, Kapoor S, Mark TE. Pediatric parameters and equipment. In: Anderson CC, Kapoor S, Mark TE, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 23rd ed. Elsevier; 2024:i-iii.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath-to-breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

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